

NAME: _____ MATRIC NUMBER: _____

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 1 SEMESTER 1 2022-2023

MS1016 – THERMODYNAMICS OF MATERIAL

20 September 2022

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
 2. Answer **ALL** questions.
 3. This is an open book test.
 4. All questions have **ONLY ONE** correct answer for the MCQ questions.
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PART I – SHORT QUESTIONS

- (1) Give a brief description of the terms, ‘reversible’ and ‘irreversible’ work in thermodynamics. Explain the effect of irreversible work on the efficiency of the system. (20 marks)

Reversible work is defined as work performed under equilibrium condition. This implies a quasi-static condition.

Given that the entropy change of the combined system and surrounding is greater than or equal to 0.

$$dS_{\text{total}} \geq 0$$

For reversible work the change in entropy of the system is 0, $dS = 0$

For an irreversible condition change in entropy of the system greater than 0, $dS > 0$

Since $dS = \delta Q/T$, this means

$$dS_{\text{irreversible}} = dS_{\text{reversible}} + \text{lost work}$$

Irreversible work results in reduced efficiency as a result of lost work caused by spontaneous increase in the entropy

2. Explain the term 'Gibbs free energy' of a system.

Explain why the Gibbs free energy is unique for an isolated system when the temperature is constant.

(20 marks)

The Gibbs free energy is defined the maximum energy in a system that is available to do work, other than mechanical work. It is defined as

$$G = H - TS$$

Where H is the enthalpy, T the temperature and S the entropy.

$$\therefore dG = dH - TdS - SdT$$

From Enthalpy equation, $dH = dU + PdV + VdP$

From first Law (system performing work),

$$dU = \delta Q - PdV$$

Substituting $dH = TdS - PdV + PdV + VdP = TdS + VdP$

Since $\delta Q = TdS$.

Therefore $dG = VdP - SdT$

At constant, T and P, dT and $dP = 0$,

Therefore $dG = 0$

This means that G is constant and therefore unique at constant temperature and pressure.

3. Using S as a function of T and V, show that.

$$dS = \frac{c_v}{T} dT + \left(\frac{dP}{dT} \right)_v dV$$

S = entropy, c_v = constant volume heat capacity, T = temperature, P = pressure, V = volume

(20 marks)

Let $S = S(T, V)$

$$\therefore dS = \left(\frac{\partial S}{\partial T} \right)_v dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

But

$$dS = \frac{\delta Q}{T} = \left(\frac{\partial Q}{\partial T} \right)_v \frac{dT}{T} = \frac{C_v dT}{T}$$

$$\therefore \left(\frac{\partial S}{\partial T} \right)_v = \frac{C_v}{T}$$

$$\therefore dS = \frac{C_v}{T} dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

From the relationship,

$$dF = -SdT - PdV$$

Maxwell's relationship gives

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_v$$

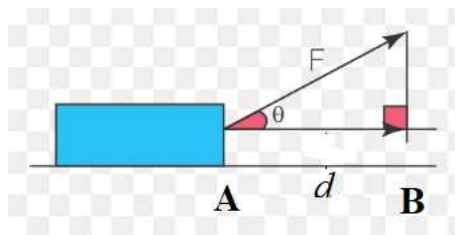
$$\therefore dS = \frac{C_v}{T} dT + \left(\frac{\partial P}{\partial T} \right)_v dV$$

PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(40 marks)

- (a) The work done is pulling the object with a force, F by a distance, d from point A to B, as shown in the figure below is



- A $(F) \times d$
- B $(F \cos \theta) \times d$
- C $(F \sin \theta) \times d$
- D $(F) \times (d \cos \theta)$
- E $(F) \times (d \sin \theta)$

ANS: B

- (b) Which of the following statement best describe an isothermal process accurately?

- A constant internal energy to maintain constant T
- B constant entropy to maintain constant T
- C no heat absorbed in the process to maintain constant T
- D no heat released in the process to maintain constant T
- E heat absorbed or released to maintain constant T

ANS: E

(c) Which of the following statements is NOT a true property of the entropy of a system?

- A cannot be decreased
- B always increase spontaneously
- C unit is Joules/K
- D $dS = 0$ for a reversible process
- E can be decreased

ANS: A

(d) When the pressure of a system is increased adiabatically.

- A temperature remains constant
- B temperature will decrease
- C temperature will increase
- D insufficient information to determine the path of temperature
- E none of the above

ANS: C

(e) The enthalpy of a solid metal represents the

- A crystal structure of the metal
- B temperature of the metal
- C the bond energy of the atoms
- D the vibrational energy of the atoms
- E All the above

ANS: E

(f) Which of the following best describes the free energy of a system?

- A energy of the system of when enthalpy and entropy are maximum
- B energy of the system available to perform mechanical work
- C total internal energy of the system
- D total energy of the system available to perform chemical work
- E total energy of the system at constant pressure

ANS: D

(g) The rate of decrease of free energy of the Gibbs of a system with respect to the temperature is defined as

- A entropy
- B internal energy
- C pressure
- D volume
- E enthalpy

ANS: A

(h) Which of the following is NOT a reversible process

- A equilibrium
- B compression of a gas
- C burning of fuel
- D expansion of a solid metal
- E compression of a spring

ANS: C

(i) The efficiency of a heat machine can be increased by,

- A decreasing the temperature of the energy input
- B minimizing the entropy increase
- C increasing the temperature of the energy input
- D minimize the temperature between the input and output energies, $\Delta T = 0$
- E minimize the temperature of the heat machine

ANS: B or C

(j) The enthalpy of formation of a compound is:

- A heat released at equilibrium oxygen pressure
- B heat released during a reaction
- C internal heat at constant pressure
- D enthalpy change of the material between 2 states
- E total internal energy of the reaction at atmospheric pressure

ANS: D